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БЧХ-кода и РС-кода

Боуз, Рой-Чоудхури, Хоквингем

Рид, Соломон → границы Синг $d \leq \quad = n-k+1$.

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2 Код с проверкой на чётности

3. код Хэмминга

4. расширенный код Хэмминга

$$H \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \quad (7,4)$$

1 $\quad x^2+1$

x $\quad x^2+x+1$

$1+x$

x^2

α^0

α

α^2

$GF(2)$ $\quad 1+x+x^3 = p(x)$ $\rightarrow GF(2^3)$

$2^3 = 8$

$0\,0\,0$

$\dots$

$1\,1\,1$

0	0
α^0	1
α^1	x
α^2	x^2
α^3	$1+x$
α^4	$x+x^2$
α^5	$1+x+x^2$
α^6	$1+x^2$.

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$H = (1\ \alpha\ \alpha^2\ \alpha^3\ \alpha^4\ \alpha^5\ \alpha^6)$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$GF(2) \to GF(2^4)$

$(15,11)$

$H = (1\ \alpha\ \alpha^2\ \alpha^3 \dots \alpha^{14})$ $\quad p(x) = 1 + x + x^4$

$c(x) + e(x) = b(x)$

$s = \sum_{i=0}^{n-1} b_i \cdot \alpha^i = \sum_{i=0}^{n-1} (c_i + e_i)\alpha^i = \sum_{i=0}^{n-1} e_i \alpha^i = e(\alpha)$

В позиции i — ошибка $\quad e(x) = x^i,\ s = \alpha^i$

$e(x) = x^i + x^j,\ i \neq j$

$$\begin{cases} S_1 = \alpha^i + \alpha^j \\ S_2 = f(\alpha^i) + f(\alpha^j) \end{cases}$$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \dots & \alpha^{14} \\ \beta_1 & \beta_2 & \beta_3 & \dots & \beta_{15} \end{pmatrix} \quad 8 \times 15.$$

$\beta_i \in GF(2^m)$

$f(\alpha) = c \cdot \alpha$ $\qquad GF(2)$

$f(\alpha) = \alpha^2$ $\qquad b(x^2) = b^2(x)$

$f(\alpha) = \alpha^3$ $\qquad 1^2 = 1 \quad 0^2 = 0$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \alpha^8 & \alpha^9 & \alpha^{10} & \alpha^{11} & \alpha^{12} & \alpha^{13} & \alpha^{14} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & \alpha^1 & \alpha^4 & \alpha^7 & \alpha^{10} & \alpha^{13} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} \end{pmatrix}$$

$x = \alpha^i \qquad y = \alpha^j$

i, j — позиции ошибки

$$\begin{cases} x + y = S_1 \\ x^3 + y^3 = S_2 \end{cases}$$

при одной или двух ошибки $S_1 \neq 0$

$\frac{x^3 + y^3}{x + y} = \frac{S_2}{S_1}$ $\qquad (x^3 + y^3) = (x+y)(x^2 + y^2 + xy)$

$(a+b)^2 = a^2 + 2ab + b^2 = a^2 + b^2.$

$x^2 + y^2 + xy = \frac{S_2}{S_1}$

$x + y = S_1 \qquad (x+y)^2 = S_1^2$

$x^2 + y^2 = S_1^2$

$S_1^2 + xy = \frac{S_2}{S_1}$ $\qquad$ т Виета.

$$\begin{cases} xy = \frac{S_2}{S_1} - S_1^2 \\ x + y = S_1 \end{cases}$$

$z^2 + \sigma_1 z + \sigma_2 = 0$

$\sigma_1 = S_1 \qquad \sigma_2 = \frac{S_2}{S_1} - S_1^2$

$GF(2) \longrightarrow GF(2^4) \quad p(x) = 1 + x + x^4$

степень α $\qquad$ полином σ $\qquad$ двоично r 0000.

0	1	0001
1	x	0010
2	x^2	0100
3	x^3	1000
4	$1+x$	0011
5	$x+x^2$	0110
6	x^2+x^3	1100
7	$1+x+x^3$	1011
8	$1+x^2$	0101
9	$x+x^3$	1010
10	$1+x+x^2$	0111
11	$x+x^2+x^3$	1110
12	$1+x+x^2+x^3$	1111
13	$1+x^2+x^3$	1101
14	$1+x^3$	1001

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \alpha^8 & \alpha^9 & \alpha^{10} & \alpha^{11} & \alpha^{12} & \alpha^{13} & \alpha^{14} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} \end{pmatrix}$$

$$H_0 = \begin{matrix} 000100 \\ 001001 \\ 010011 \\ 100010 \end{matrix} \rightarrow$$

$$H = \begin{matrix} 011110 \\ 001010 \\ 000110 \\ 100011 \end{matrix} \rightarrow$$

111 00 11 000 11 011

11 0 00 11 00 1 11 011

$s_1 = \alpha^2 + \alpha^9 = \alpha^{11}$ $\quad s_2 = \alpha^6 + \alpha^{12} = \alpha^4$

$= \sigma^2$

$$\begin{cases} xy = \dfrac{s_2}{s_1} - s_1^2 \\ x+y = s_1 = \alpha^1 \end{cases}$$

$$\frac{\alpha^4}{\alpha^{11}} + (\alpha^{11})^2 = \alpha^8 + \alpha^7 = \alpha^{11}$$

$$= \frac{1}{\alpha^7} + \alpha^7 = \frac{1+\alpha^{14}}{\alpha^7} = \frac{\alpha^5}{\alpha^7} = \alpha^{-4} = \alpha^{11}.$$

$\sigma^2 = \alpha^{11} \quad \alpha^1 = \alpha^{11}$

$z^2 + \alpha^{11} \cdot z + \alpha^{11} = \emptyset$

$(\alpha^1)^2 + \alpha^{11} \cdot \alpha^2 + \alpha^{11} = \alpha^4 + \alpha^{13} + \alpha^{11} = \alpha^2 \quad \alpha^9$

$1+x+1+x^2+x^3+x+x^2+x^3$

$(\alpha^9)^2 + \alpha^{11}\alpha^9 + \alpha^{11} = \alpha^3 + \alpha^5 + \alpha^{11} = \emptyset$

$x^3+x+x^2+x+x^2+x^3 = \emptyset$